

# Adaptive Heuristic Portfolio TSP Report

## Overview

- Condition count: 8.
- Best held-out TSPLIB gap: phase8\_best\_single\_fixed\_heuristic.
- Best combined transfer gap: phase8\_oracle\_selector.
- Lowest selector regret: phase8\_best\_single\_fixed\_heuristic.
- Pareto-efficient conditions: phase8\_best\_single\_fixed\_heuristic, phase8\_full\_solver\_evolution, phase8\_llm\_evolved\_adaptive\_controller, phase8\_llm\_static\_selector, phase8\_random\_portfolio.

## Run Metadata

- run\_name: run\_20260520\_182519\_e
- started\_at\_local: 2026-05-20 18:25:19
- finished\_at\_local: 2026-05-20 18:55:09
- duration\_hhmm: 00:30
- duration\_seconds: 1790.296
- seed\_offset: 4000
- replicate\_label: e
- judge\_status: enabled
- judge\_provider: openai
- judge\_model: gpt-4.1-mini

## Condition Comparison

| Condition                          | Mode             | Replay | Final TSPLIB Gap | Selector Regret | Runtime (ms) | Runtime-Adjusted Gap | Family Gap | Transfer Gap | Mean Novelty | Mean Complexity | Pareto Efficient |
|------------------------------------|------------------|--------|------------------|-----------------|--------------|----------------------|------------|--------------|--------------|-----------------|------------------|
| phase8_best_single_fixed_heuristic | best_fixed       | none   | 0.07226          | 0.0             | 1478.632943  | 0.07226              | 0.007284   | 0.03122      | 0.0          | 0.0             | True             |
| phase8_random_portfolio            | random_portfolio | none   | 0.238028         | 0.165768        | 204.906786   | 0.238028             | 0.06063    | 0.125629     | 0.0          | 0.0             | True             |
| phase8_oracle_selector             | oracle_selector  | none   | 0.07226          | 0.0             | 1515.764186  | 0.074075             | 8.9e-05    | 0.026678     | 0.0          | 0.0             | False            |

| Condition                                              | Mode                | Replay            | Final TSPLI B Gap | Selector Regret | Runtime (ms) | Runtime -Adjusted Gap | Family Gap | Transfer Gap | Mean Novelty | Mean Complexity | Pareto Efficient |
|--------------------------------------------------------|---------------------|-------------------|-------------------|-----------------|--------------|-----------------------|------------|--------------|--------------|-----------------|------------------|
| phase8_supervised_ml_selector                          | supervised_selector | none              | 0.07226           | 0.0             | 1622.641086  | 0.079298              | 0.007284   | 0.03122      | 0.0          | 0.0             | False            |
| phase8_llm_static_selector                             | static_selector     | none              | 0.229271          | 0.157011        | 265.383486   | 0.229271              | 0.045605   | 0.113271     | 0.0          | 0.62            | True             |
| phase8_llm_evolved_adaptive_controller                 | adaptive_controller | none              | 0.214311          | 0.142051        | 506.734      | 0.214311              | 0.02333    | 0.093693     | 0.426878     | 0.62            | True             |
| phase8_llm_evolved_controller_diversity_failure_replay | adaptive_controller | diversity_failure | 0.203034          | 0.130774        | 302.534086   | 0.203034              | 0.063338   | 0.114805     | 0.811584     | 1.19            | False            |
| phase8_full_solver_evolution                           | full_solver         | none              | 0.165894          | 0.093634        | 301.039157   | 0.165894              | 0.010006   | 0.067438     | 0.0          | 0.76            | True             |

## Condition Notes

### phase8\_best\_single\_fixed\_heuristic

- Execution mode best\_fixed with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.07226.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 1478.632943 ms, runtime inflation 0.0, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: True.
- Fixed heuristic choice: farthest\_insertion\_two\_opt.

### phase8\_random\_portfolio

- Execution mode random\_portfolio with replay none.
- Final held-out gap 0.238028, selector regret 0.165768, runtime-adjusted gap 0.238028.
- Family transfer 0.060063 and combined transfer 0.125629.
- Runtime 204.906786 ms, runtime inflation -0.861421, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: True.

### phase8\_oracle\_selector

- Execution mode oracle\_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.074075.

- Family transfer 8.9e-05 and combined transfer 0.026678.
- Runtime 1515.764186 ms, runtime inflation 0.025112, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

### **phase8\_supervised\_ml\_selector**

- Execution mode supervised\_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.079298.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 1622.641086 ms, runtime inflation 0.097393, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

### **phase8\_llm\_static\_selector**

- Execution mode static\_selector with replay none.
- Final held-out gap 0.229271, selector regret 0.157011, runtime-adjusted gap 0.229271.
- Family transfer 0.045605 and combined transfer 0.113271.
- Runtime 265.383486 ms, runtime inflation -0.820521, mean novelty 0.0, and complexity 0.62.
- Pareto efficient: True.
- Controller signature: annealed\_multi\_start:cluster\_first\_local\_search:farthest\_insertion\_two\_opt:annealed\_multi\_start:stag3:cand2:restart1.
- Controller rule summary: {"acceptance\_bias": 0.005, "candidate\_limit\_offset": 2, "clustered\_heuristic": "cluster\_first\_local\_search", "corridor\_heuristic": "edge\_preserving\_restart", "default\_heuristic": "farthest\_insertion\_two\_opt", "description": "Deterministic instance-conditioned frozen-portfolio selector targeting farthest-insertion + 2-opt for general performance, with structure-aware fallbacks for clustered/grid/bottleneck/corridor and nearest-neighbor traps. Heuristic-specific schedule tuning uses", "failed\_perturbation\_threshold": 2, "grid\_like\_heuristic": "farthest\_insertion\_two\_opt", "low\_improvement\_threshold": 0.08, "name": "det\_tsp\_hh\_oracle\_portfolio\_like\_v1", "nearest\_neighbor\_trap\_heuristic": "candidate\_pruned\_two\_opt", "random\_like\_heuristic": "annealed\_multi\_start", "restart\_offset": 1, "stagnation\_switch\_heuristic": "annealed\_multi\_start", "stagnation\_threshold": 3, "temperature\_scale": 1.1, "time\_budget\_trigger": 0.55, "two\_cluster\_bottleneck\_heuristic": "farthest\_insertion\_two\_opt"}

### **phase8\_llm\_evolved\_adaptive\_controller**

- Execution mode adaptive\_controller with replay none.
- Final held-out gap 0.214311, selector regret 0.142051, runtime-adjusted gap 0.214311.
- Family transfer 0.023333 and combined transfer 0.093693.
- Runtime 506.734 ms, runtime inflation -0.657296, mean novelty 0.426878, and complexity 0.62.
- Pareto efficient: True.
- Controller signature: annealed\_multi\_start:farthest\_insertion\_two\_opt:farthest\_insertion\_two\_opt:nearest\_neighbor\_multistart:stag3:cand2:restart1.
- Controller rule summary: {"acceptance\_bias": 0.008, "candidate\_limit\_offset": 2, "clustered\_heuristic": "farthest\_insertion\_two\_opt", "corridor\_heuristic": "farthest\_insertion\_two\_opt", "default\_heuristic": "farthest\_insertion\_two\_opt", "description": "Deterministic instance-adaptive portfolio router with stagnation-triggered switching and bounded schedule tuning over the frozen heuristic set.", "failed\_perturbation\_threshold": 3, "grid\_like\_heuristic": "farthest\_insertion\_two\_opt", "low\_improvement\_threshold": 0.1, "name":

"tsp\_hh\_controller\_deterministic\_structure\_switching\_v3", "nearest\_neighbor\_trap\_heuristic": "annealed\_multi\_start", "random\_like\_heuristic": "annealed\_multi\_start", "restart\_offset": 1, "stagnation\_switch\_heuristic": "nearest\_neighbor\_multistart", "stagnation\_threshold": 3, "temperature\_scale": 1.2, "time\_budget\_trigger": 0.7, "two\_cluster\_bottleneck\_heuristic": "farthest\_insertion\_two\_opt"}

### **phase8\_llm\_evolved\_controller\_diversity\_failure\_replay**

- Execution mode adaptive\_controller with replay diversity\_failure.
- Final held-out gap 0.203034, selector regret 0.130774, runtime-adjusted gap 0.203034.
- Family transfer 0.063338 and combined transfer 0.114805.
- Runtime 302.534086 ms, runtime inflation -0.795396, mean novelty 0.811584, and complexity 1.19.
- Pareto efficient: False.
- Controller signature: nearest\_neighbor\_multistart:cluster\_first\_local\_search:edge\_preserving\_restart:limited\_three\_opt:stag4:cand-1:restart2.
- Controller rule summary: {"acceptance\_bias": 0.004, "candidate\_limit\_offset": -1, "clustered\_heuristic": "cluster\_first\_local\_search", "corridor\_heuristic": "candidate\_pruned\_two\_opt", "default\_heuristic": "farthest\_insertion\_two\_opt", "description": "Deterministic instance-adaptive frozen-portfolio controller using structure-aware heuristic role mapping and a conservative stagnation switch to avoid diversity-collapse during replay.", "failed\_perturbation\_threshold": 3, "grid\_like\_heuristic": "cheapest\_insertion\_two\_opt", "low\_improvement\_threshold": 0.12, "name": "tsp\_hh\_frozen\_portfolio\_v4\_deterministic\_instance\_adaptive", "nearest\_neighbor\_trap\_heuristic": "candidate\_pruned\_two\_opt", "random\_like\_heuristic": "nearest\_neighbor\_multistart", "restart\_offset": 2, "stagnation\_switch\_heuristic": "limited\_three\_opt", "stagnation\_threshold": 4, "temperature\_scale": 1.25, "time\_budget\_trigger": 0.68, "two\_cluster\_bottleneck\_heuristic": "edge\_preserving\_restart"}

### **phase8\_full\_solver\_evolution**

- Execution mode full\_solver with replay none.
- Final held-out gap 0.165894, selector regret 0.093634, runtime-adjusted gap 0.165894.
- Family transfer 0.010006 and combined transfer 0.067438.
- Runtime 301.039157 ms, runtime inflation -0.796407, mean novelty 0.0, and complexity 0.76.
- Pareto efficient: True.

## **Judge Appendix**

### **Conservative Interpretation of TSP Adaptive Heuristic Portfolio Suite Results**

#### Primary Endpoint: Held-Out TSPLIB Gap

- **Best performance:**
- *phase8\_best\_single\_fixed\_heuristic* achieves the lowest held-out TSPLIB gap (0.07226).
- *phase8\_oracle\_selector* matches this TSPLIB gap (0.07226) but is not deployable (oracle upper bound).
- **Adaptive & static selectors:**
- *phase8\_llm\_evolved\_adaptive\_controller* shows moderately higher TSPLIB gap (0.21431), about 3x worse than the best fixed heuristic.
- *phase8\_llm\_static\_selector* has even higher gap (0.22927).
- *phase8\_full\_solver\_evolution* intermediate at (0.16589).

- **Random portfolio** is substantially worse (0.23803).

#### #### Selector Regret vs Oracle Portfolio

- Oracle selector has zero regret (by definition).
- **Best single fixed heuristic** has zero regret (it is fixed).
- Adaptive controllers and static selector incur positive selector regret (~0.13–0.16), indicating imperfect adaptivity relative to oracle.
- **phase8\_full\_solver\_evolution** shows lower regret (0.0936) than adaptive controllers but worse gap than fixed heuristic.

#### #### Runtime-Adjusted Gap and Runtime Inflation

- The best single fixed heuristic has runtime-adjusted gap 0.07226 with **no runtime inflation** (baseline).
- Oracle selector's runtime inflation is marginal (2.5%), maintaining similar runtime-adjusted gap (~0.074).
- Adaptive controller (**phase8\_llm\_evolved\_adaptive\_controller**), while having higher gap, reduces runtime compared to baseline (~65.7% runtime inflation), resulting in moderate runtime-adjusted gap (0.2143).
- Static selector and random portfolio get lower runtime (negative runtime inflation >80%) but at cost of much worse gap (~0.23 and 0.24).
- **phase8\_full\_solver\_evolution** balances runtime inflation (-79.6%) with moderate gap (0.1659).

#### #### Pareto Efficiency

- Pareto-efficient conditions include:
- **phase8\_best\_single\_fixed\_heuristic** (best gap, zero inflation)
- **phase8\_full\_solver\_evolution** (good gap/runtime tradeoff)
- **phase8\_llm\_evolved\_adaptive\_controller** (adaptive, runtime savings)
- **phase8\_llm\_static\_selector** (static selector, runtime savings at cost of gap)
- **phase8\_random\_portfolio** (runtime efficient but poor gap)
- Oracle selector is **not Pareto efficient**, likely due to higher runtime inflation without gap improvement over best fixed heuristic.

#### #### Cross-Family Transfer Gap

- Transfer gaps (generalization from training families to held-out TSPLIB) are smallest for:
- Oracle selector (0.0267)
- Best single fixed heuristic (0.0312)
- Adaptive controllers exhibit higher transfer gaps (~0.09–0.11) indicating less robust cross-family generalization.
- Static and random selectors have highest transfer gaps (0.11+).

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## Summary and Recommendations

- The **best single fixed heuristic (farthest\_insertion\_two\_opt)** remains a strong baseline with best held-out TSPLIB gap (0.07226), zero selector regret, no runtime inflation, and good transfer gap; it is Pareto efficient.

- The **oracle selector**, while an idealized upper bound, matches this gap with slight runtime inflation and thus serves as a useful performance ceiling.
- **Adaptive controllers** (`phase8_llm_evolved_adaptive_controller`) achieve meaningful runtime reduction (>60% runtime savings) but with increased TSPLIB gap (~0.21) and moderate selector regret (~0.14). This indicates runtime-performance tradeoffs in instance-adaptive hyper-heuristic control.
- The **static selector** achieves large runtime gains but at considerable accuracy loss (gap ~0.23) and selector regret (~0.16), limiting interpretability and practical appeal.
- **Cross-family transfer robustness** is best approached by fixed heuristics and oracle selectors, while adaptive controllers lag behind, suggesting limitations in generalization.
- No condition unambiguously outperforms the best single fixed heuristic on the primary endpoint without runtime cost inflation.
- Thus, current results support emphasizing **interpretable instance-adaptive control** that approaches the oracle/benchmark gap without excessive runtime inflation, rather than claims of discovering new heuristics.
- Future work might focus on narrowing the gap of adaptive selectors while preserving their runtime efficiency and cross-family robustness.

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### Concise key points:

| Condition                               | TSP LIB Gap        | Runtime-A djusted Gap | Runtime Inflation | Selecto r Regret | Trans fer Gap | Pareto Efficient | Notes                                            |
|-----------------------------------------|--------------------|-----------------------|-------------------|------------------|---------------|------------------|--------------------------------------------------|
| phase8_best_single_fixe d_heuristic     | <b>**0.07226**</b> | 0.07226               | 0.0               | 0.0              | 0.0312        | Yes              | Strong baseline, no runtime cost                 |
| phase8_oracle_selector                  | 0.07226            | 0.07407               | +2.5%             | 0.0              | 0.0267        | No               | Ideal upper bound, not deployable                |
| phase8_llm_evolved_ad aptive_controller | 0.21431            | 0.21431               | -65.7%            | 0.142            | 0.0937        | Yes              | Runtime-efficient adaptive control, gap tradeoff |
| phase8_full_solver_evol ution           | 0.16589            | 0.16589               | -79.6%            | 0.094            | 0.0674        | Yes              | Balanced runtime/performance                     |
| phase8_llm_static_select or             | 0.22927            | 0.22927               | -82.1%            | 0.157            | 0.1133        | Yes              | Static selector, runtime gain but gap loss       |
| phase8_random_portfoli o                | 0.23803            | 0.23803               | -86.1%            | 0.166            | 0.126         | Yes              | Poor gap, no adaptation                          |

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### **\*\*Conclusion:\*\***

The best fixed heuristic remains the conservative choice for TSP on TSPLIB benchmarks. Instance-adaptive hyper-heuristic control shows promise in runtime reduction but requires further tuning to close the performance gap without sacrificing interpretability or transfer robustness.